FRE 7841 - HEDGE FUND STRATEGIES

CONVERTIBLE BOND ARBITRAGE

A systematic, dynamically delta-hedged strategy with residual gamma exposure, exploiting mispricings between model fair value and market prices of convertible bonds.

PERFORMANCE

+6.21%	0.43	−1.44%
Annualised Return	Sharpe Ratio	Max Drawdown

RISK-ADJUSTED // MARKET COMPARISON

+2.85%	−0.07	+2.83%	−0.10
Alpha vs S&P 500	Beta vs S&P 500	Alpha vs 60/40 Portfolio	Beta vs 60/40 Portfolio

CORRELATION // TRADE STATISTICS

−0.27	−0.26	56.4%	$2.08M
Corr vs S&P 500	Corr vs 60/40 Portfolio	Win Rate	Net P&L (3 Yrs)

UNIVERSE	10 convertible bonds - REITs, Utilities & Tech
SIGNAL	Bond floor + CRR binomial model fair value gap
HEDGE	Threshold delta rebalancing: long bond, short stock
PERIOD	Jan 2023 – Feb 2026 | $10M starting capital

Adding this strategy to a portfolio reduces risk, not adds it. Negative correlations (−0.27, −0.26) mean it earns precisely when traditional allocations are hurting.

Darshit Sarda (ds8286) Jinay Jain (jj4407)

HOW THE STRATEGY WORKS

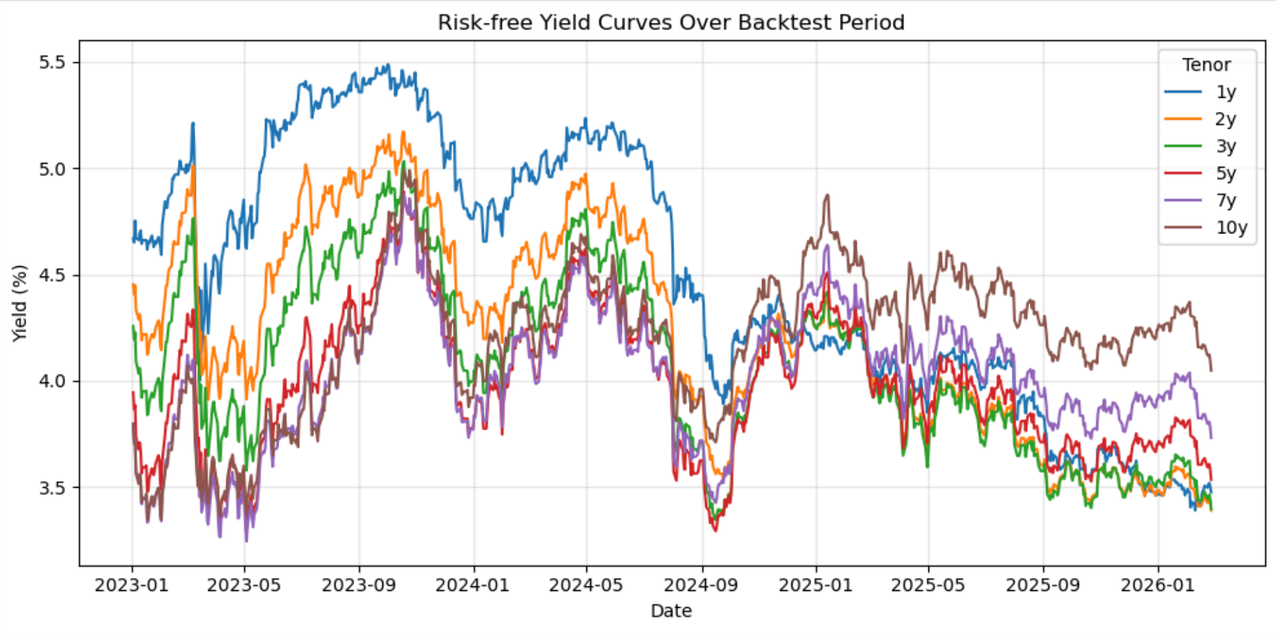
- Instrument: Plain vanilla convertible bonds = straight bond + embedded equity call option
- Fair Value: Bond Floor (PV of coupons + par, discounted at ZERO COUPON BONDS + BBB spread) + Option Value (CRR binomial tree, 100 steps, American call)
- Signal: Mispricing Gap = Fair Value - Market Clean Price (% of par)
- Hedge: Long bond, short delta-equivalent shares. Rebalanced on threshold drift or large stock moves.
- Risk controls: $1M per position, convergence exit, 365-day max hold, stop-loss. Uninvested cash earns T-bill rate.
- Return sources: Coupon accrual, price convergence, gamma from rebalancing, short rebate. transaction costs are explicitly deducted.

NOTE

Delta hedging chosen over CDS and duration variants - CDS markets illiquid for BBB and BB rated bonds; avg position delta of 0.55 confirms equity risk dominates over duration risk, validating hedge choice

DISCOUNT RATE METHODOLOGY

Component	Discount Rate	Rationale
Bond Floor	ZC Treasury + BBB credit spread	Captures straight bond default risk
Equity Option	ZC Treasury only	Risk-neutral pricing - no credit adjustment



- Zero-coupon Treasury rates downloaded from Bloomberg - theoretically correct for discounting individual cash flows at their own maturity-matched rate
- Bonds have non-benchmark maturities (e.g. 2.3, 3.7 years) - linear interpolation between adjacent tenors
- Generic BBB spread used for bond floor - issuer-specific CDS would be more precise but unavailable.

SCREENING FROM 19 TO 10 BONDS

Initial Screen (Bloomberg) → 19 bonds:

Filter	Criteria
Market	US Convertible Bonds,
Issue Size	≥ $200M outstanding
Credit Rating	BB- to BBB+
Bond Status	Active only
Issue Date	On or after 2019
Maturity	Before 2030

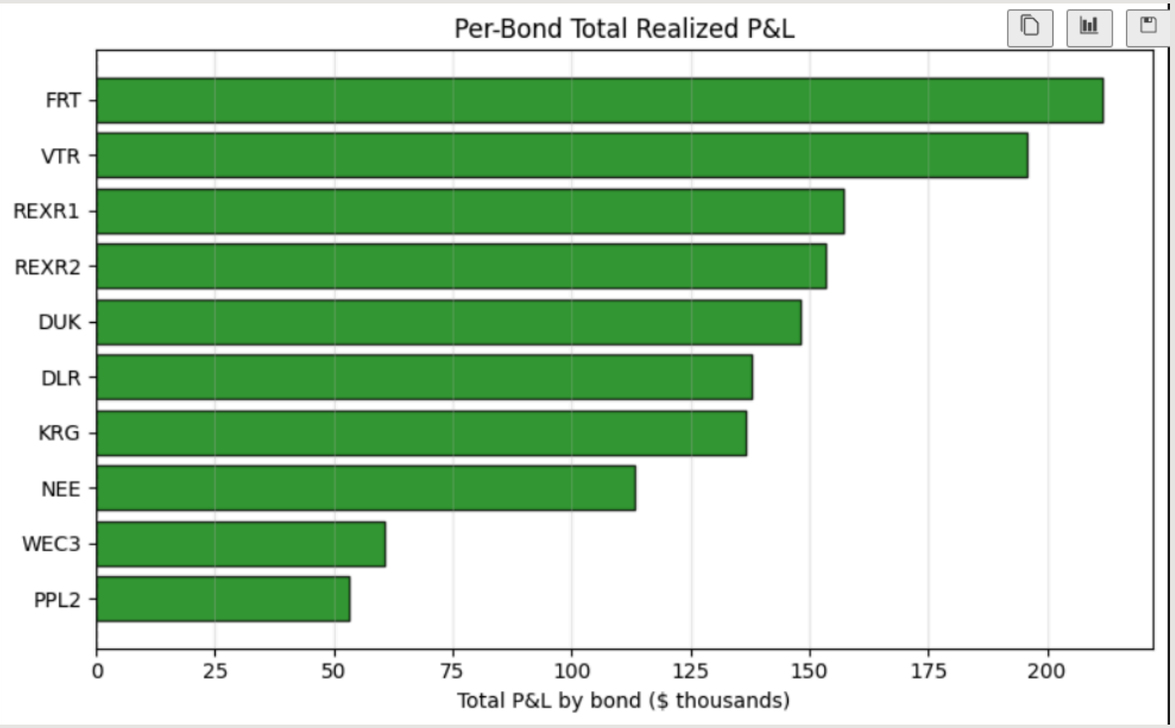
Reduced to 10 bonds for trading:

- All 19 bonds are plain vanilla structures - no contingent or mandatory conversion features
- 9 bonds excluded: 1-year implied volatility (only tenor available without full vol surface) systematically overstated option value for long-dated bonds, producing persistent unrealistic mispricing gaps that never converged
- Generic BBB credit spread could not fully capture issuer-specific risk for excluded names
- Final universe: DLR, DUK, FRT, KRG, NEE, PPL2, REXR1, REXR2, VTR, WEC3

PORTFOLIO CONSTRUCTION & PERFORMANCE ATTRIBUTION BY BOND

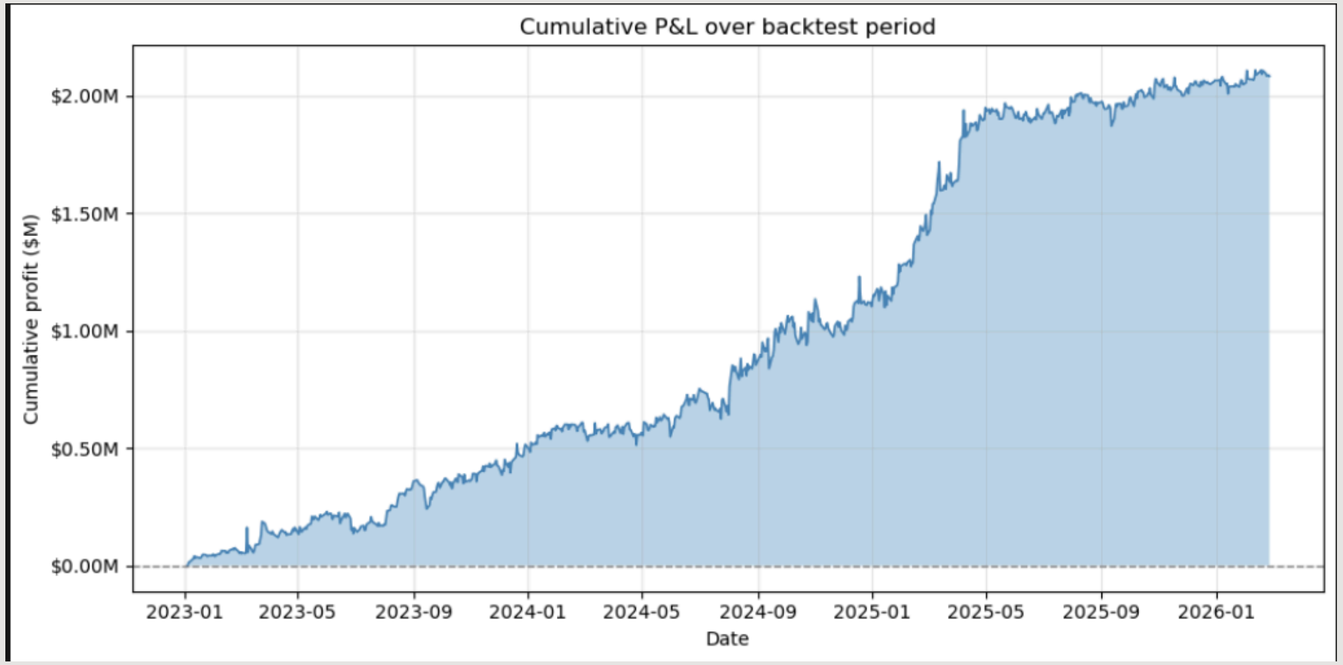
CAPITAL STRUCTURE:

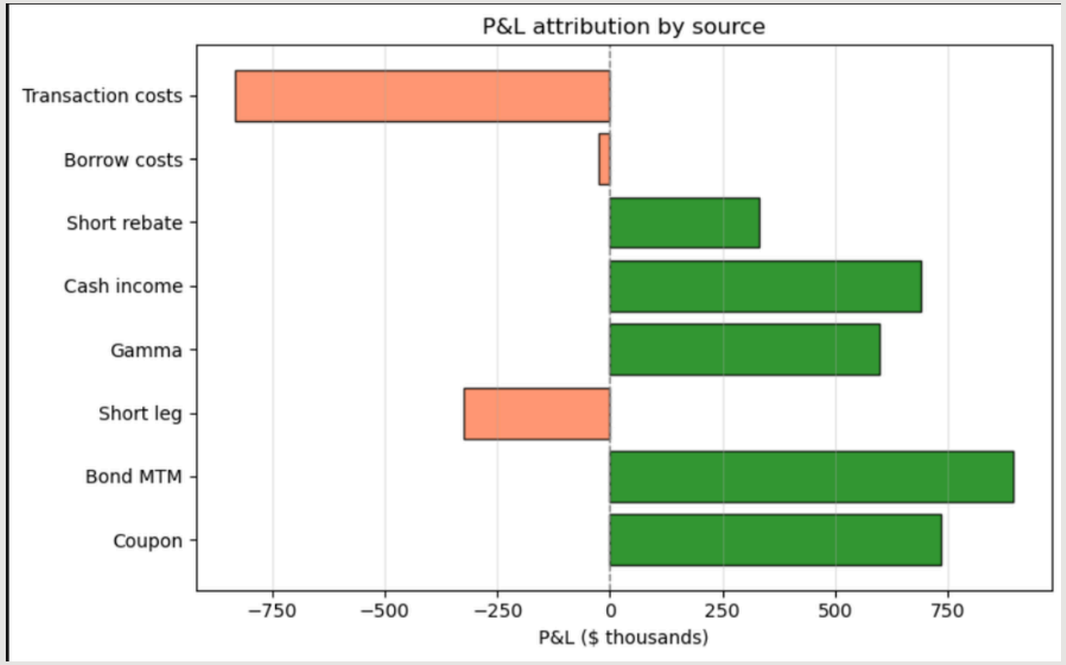
- $10M CAPITAL | $1M FIXED ALLOCATION PER BOND | MAX 10 SIMULTANEOUS POSITIONS
- AVG 6 POSITIONS OPEN SIMULTANEOUSLY | 61.5% AVG CAPITAL DEPLOYED AT ANY TIME OVER THE BACKTEST | UNINVESTED CASH EARNS RISK-FREE RATE



- ALL 10 BONDS PROFITABLE - NO SINGLE NAME DRAGGED DOWN PORTFOLIO RETURNS
- FRT AND VTR TOP CONTRIBUTORS ($187K, $195K) DRIVEN BY LARGE INDIVIDUAL CONVERGENCE PAYOFFS

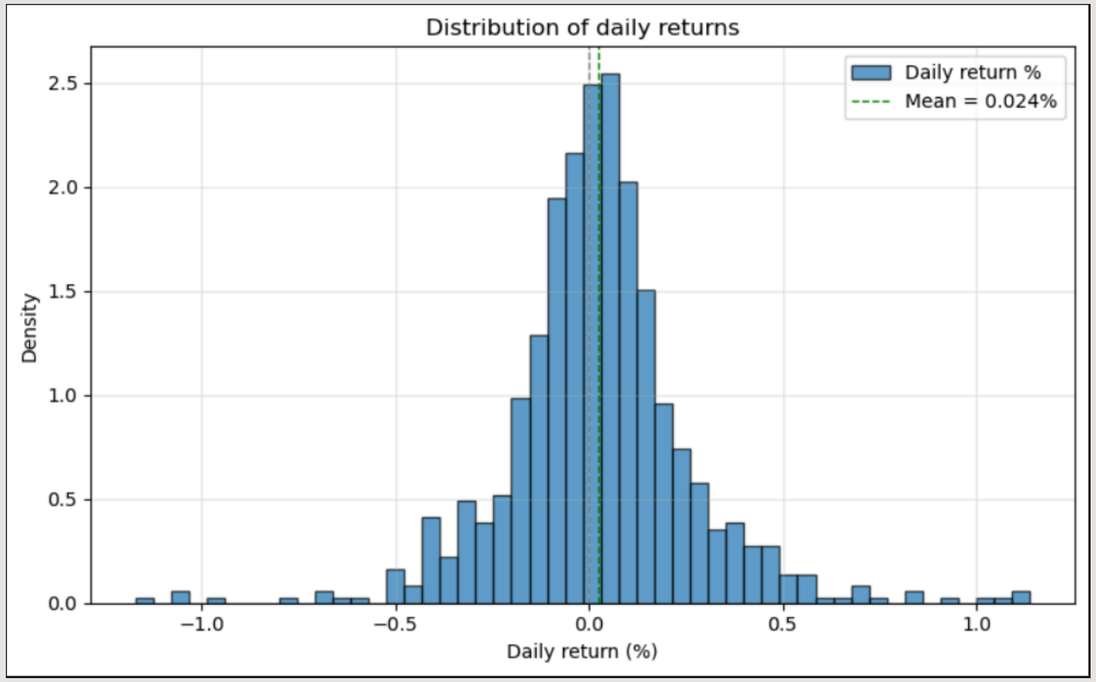
CUMULATIVE P&L





- $2.08M net profit on $10M capital over 3 years - portfolio never dipped below zero
- Acceleration in early 2025 driven by REXR1, FRT, and DUK convergence trades; plateau in late 2025 as bonds approach maturity
- Returns diversified across four independent sources - coupon ($735K), bond MTM ($895K), gamma ($598K), cash income ($690K) - not a single directional bet
- Transaction costs (−$833K) are the largest drag - 75 bps on bond notional + 10 bps on stock per entry and exit across 78 trades
- short leg loss (−$324K) nearly fully offset by short rebate (+$330K)

RETURN DISTRIBUTION & TRADE STATISTICS



- Daily NAV returns tightly clustered around zero
- Ann. volatility of 3.79% reflects residual basis risk - credit spread and rate movements - after equity risk is stripped out by delta hedging; far below typical equity strategy volatility

Trade Statistics:

- 78 total trades | Profit factor: 3.05
- Avg win: $41,917 | Avg loss: −$13,758 | Win/loss ratio: 3:1
- Avg holding period: 89 days | Avg 6 positions open simultaneously

CONCLUSIONS

Strategy delivers consistent risk-adjusted returns with minimal drawdown:

- +20.82% total return | +6.21% annualized | Sharpe 0.43 | Max drawdown −1.44% over 3.1 years
- Portfolio never went negative - strategy profitable in every sub-period

Return profile is genuinely diversified - not a single factor bet:

- Four independent P&L sources: coupon ($735K) + bond MTM ($895K) + gamma ($598K) + cash income ($690K)
- All 10 bonds profitable - no single name drives results
- Delta hedge successfully strips equity market exposure - ann. volatility 3.79%

What would improve the strategy:

- Full implied volatility surface per underlying - would restore excluded bonds and improve option pricing accuracy
- Issuer-specific CDS spreads - would replace generic BBB spread with true credit risk

BENCHMARK-INDEPENDENT ALPHA: +2.85% VS S&P 500, +2.83% VS 60/40 — WITH NEGATIVE BETA TO BOTH. THE STRATEGY DOES NOT COMPETE WITH A PORTFOLIO. IT COMPLETES ONE.

APPENDIX - SUBMITTED MATERIALS

Data

- Bond static & dynamic prices: data/raw/bond_static/CONV_BOND_DATA.xlsx, data/raw/bond_dynamic/*.xlsx
- Equity & implied vol: data/raw/stock/*_equity.xlsx
- Yield curve & discount rates: data/clean/yield_curve_clean.csv, data/clean/final_discount_rates_daily.csv
- Model outputs (per bond): data/clean/option_values/, data/clean/bond_floors/, data/clean/fair_value/
- Benchmark returns: code/sp500_and_6040_daily_returns_20230101_20260227.csv
- Backtest outputs: data/clean/backtest/ — portfolio_nav.csv, pnl_daily.csv, trades.csv, performance.csv

Code

- Pipeline: yield_curve.ipynb → bond_floor_calculation.ipynb → option_pricer.ipynb → fair_price_calculation.ipynb
- Strategy & attribution: bond_backtest.ipynb
- Benchmarks: build_6040_and_sp500_returns_fixed.py
- GitHub: github.com/DarshitSarda/Convertible_Bond_Arbitrage_Strategy

References

- Hutchinson & Gallagher (2005). Convertible Bond Arbitrage. EFMA Annual Meeting Paper; Empirical validation of delta-neutral convertible arb returns and risk decomposition.
- Agarwal, Fung, Loon & Naik (2011). Risk and Return in Convertible Arbitrage: Evidence from the Convertible Bond Market. Journal of Empirical Finance; Factor model framework for convertible arb P&L attribution; directly informs our return decomposition methodology.